Module 9: Microbiology

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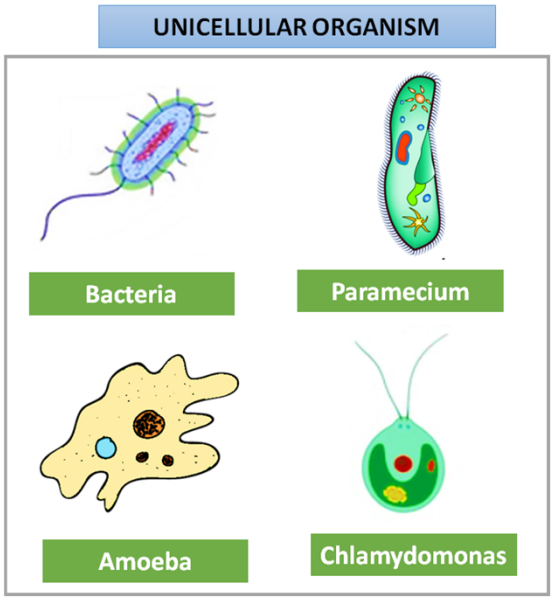
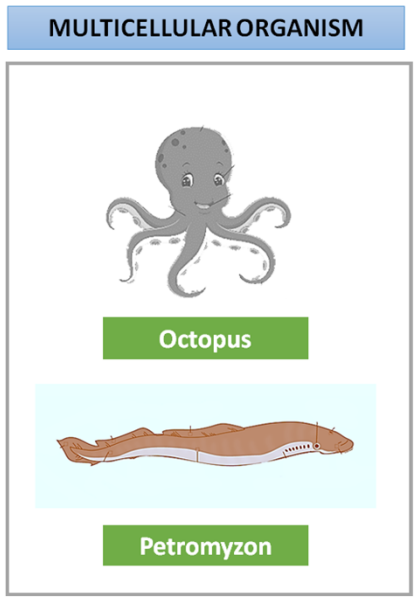
Introduction

- Microbiology is the branch of science that deals with the study of microbes.
- Most microorganisms are made up of a single cell and exhibit properties shared by all biological systems. These properties include reproduction, metabolic activity, growth, irritability, adaptability, mutation, and organization.
- **Characteristics of Unicellular organisms:**
- Unicellular organisms generally reproduce through asexual means.
- They can be either prokaryotes or eukaryotes.
- They can be found in almost any environment, from hot springs to frozen tundra.
- For movement, they have flagella, pilli or fimbrae.
- Diffusion allows nutrients to enter and exit the cell.





Unicellular organisms

- There are two major types of cells: **prokaryotic** and **eukaryotic**. Bacteria and Archaea are single-celled creatures categorized as prokaryotes (**pro = before**; **karyon– = nucleus**). Animal cells, plant cells, fungi, and protists are eukaryotes (**eu = true**).
- **Prokaryotes**
- Cell is a simple, unicellular organism with **no nucleus** or other membrane-bound organelles. Prokaryotic DNA is in the center of the cell, in a dark area known as the nucleoid. In contrast to Archaea and eukaryotes, bacteria have a cell wall constructed of **peptidoglycan** composed of carbohydrates and amino acids and a polysaccharide capsule. The cell wall provides an additional layer of protection, aids in maintaining the cell's shape, and prevents dehydration. The cell's ability to adhere to surfaces in its surroundings is enabled by its capsule. Some prokaryotes have **flagella**, **pili**, or **fimbriae**. Flagella are utilized for movement, while most pili are employed to exchange genetic material during conjugation, a sort of reproduction.
- **Eukaryotes**
- Eukaryotic cells have numerous compartments with specialized functions, neatly separated by **membrane layers**. This structure allows each compartment to maintain its own conditions, those required to perform its function. **Lysosomes**, which operate as recycling centers for the cell, must, for example, maintain an acidic pH to dispose of cellular waste. Similarly, structures known as **peroxisomes** carry out oxidation reactions and produce hydrogen peroxide, both of which would harm the cell if not properly kept away in their own "chamber." Because eukaryotic cells can sustain several habitats within a single cell, they can perform complicated metabolic reactions that prokaryotes cannot. Indeed, it is a major reason why eukaryotic cells can grow many times larger than bacterial cells

Multicellular organisms

- A multicellular organism is one made of **several cells**.
- Multicellular organisms grow through **cellular specialization** and **labor division**. Cells become specialized in one process and are **interdependent** on one another to carry out other functions. For an organism to operate properly, all its cells must work together in harmony.
- Cells of multicellular animals may also differ in appearance depending on the organelles required within the cell.
- For example: Muscle cells have more mitochondria than most other cells, allowing themto manufacture energy for movement easily.
- Pancreatic cells must produce numerous proteins and have more ribosomes and rough endoplasmic reticula to meet this requirement.

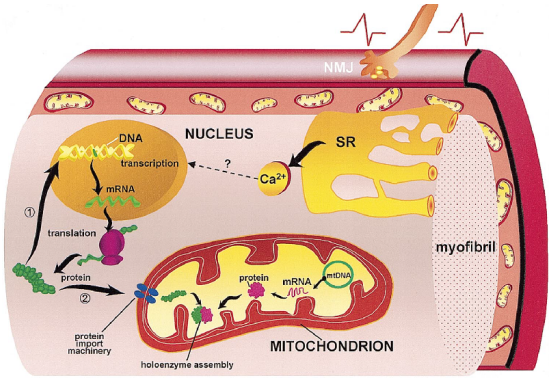


Fig: Overall synopsis of mitochondrial biogenesis in a muscle cell

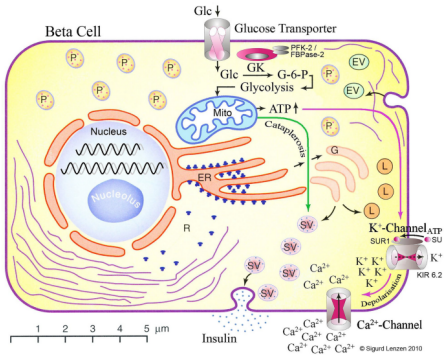
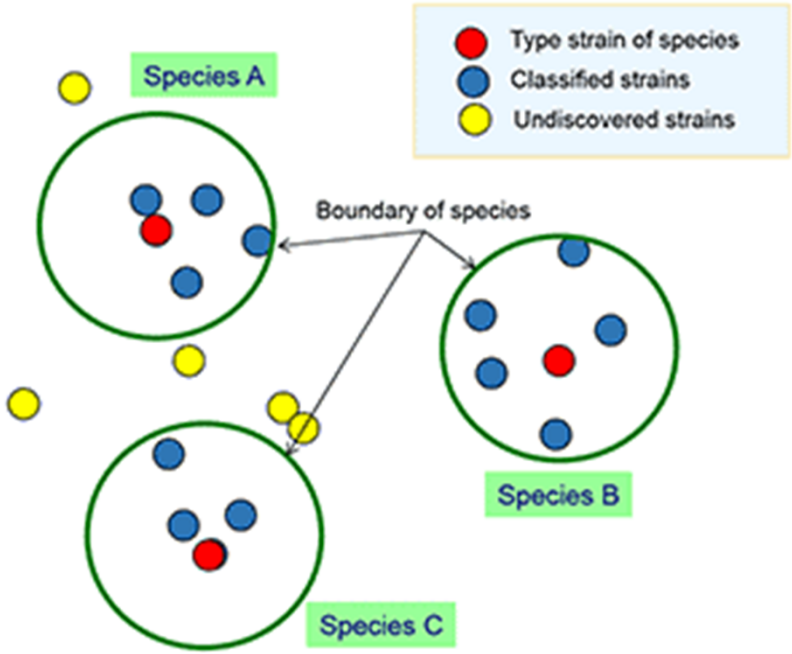


Fig: A scheme of the cellular structures of pancreatic beta cells.

Concepts of Species & Strain

- Species are the **largest group of organisms** in which any two individuals of appropriate sexes can produce **fertile offspring** through sexual reproduction.
- Species are also a **fundamental unit of classification**. Species can also be defined by their similar DNA sequence, morphology, karyotype, behavior, ecological niche, or sequential development pattern.
- A strain is a **subtype** or **genetic variant of a biological species**.
- Strains in microorganisms typically **originate from a single-cell colony of microorganisms**.
- Strains are commonly found in virology, botany, insects, and experimental rodents and are often considered an inherently artificial concept because it describes a specific intent, such as genetic isolation.



Identification of Microorganisms

- The identification of microorganisms is primarily dependent on phenotypic characteristics. However, advanced genotypic methods such as fingerprint and sequence-based techniques have recently been used.
- **Microscopy**: Morphology and staining reactions of organisms are used to classify unknown species. Gram staining distinguishes gram-positive and negative microorganisms.
- **Cultural characteristics**: The appearance of colonial growth on nutrient agar is distinctive. Diameter, outline, elevation, translucency (clear, translucent, opaque), and color are noted.
- **Biochemical characteristics**: Species that lack morphological and cultural distinctions may have metabolic differences. When glucose, lactose, sucrose, mannitol, etc., are the sole carbon supply, the organism's ability to create acidic and gaseous end-products is tested.
- In microscopy, **resolution** refers to a microscope's ability to identify fine details. This is the minimal distance between two points of a specimen that may be distinguished as different entities, either by the microscope camera or by the observer.

Microscopy: Types & Differences

Sl No	Criteria	Simple Microscope	Compound Microscope
1	Number of Lenses	Single lens system	Multiple lens system (objective and eyepiece lenses)
2	Magnification	Limited magnification (up to 300x)	Higher magnification (up to 2000x or more)
3	Lens Arrangement	Single convex lens	Combination of objective and eyepiece lenses
4	Field of View	Larger field of view	Smaller field of view
5	Resolution	Lower resolution	Higher resolution
6	Image Quality	Lower image quality compared to compound microscopes	Higher image quality due to multiple lenses
7	Sample Thickness	Suitable for observing thin samples	Suitable for observing thicker samples
8	Illumination	Requires external light source	Often includes built-in illumination systems
9	Applications	Suitable for basic observation tasks	Used in various scientific and research applications
10	Portability	Generally more portable and lightweight	Tends to be bulkier and less portable

Sl No	Criteria	Light Microscope	Electron Microscope
1	Illuminating source	Uses light (approx wavelength 400-700 nm) to illuminate the objects under view.	Uses a beam of electrons (approx equivalent wavelength 1 nm) to make objects larger for a detailed view.
2	Principle	The image is formed by the absorption of light waves.	The image is formed by scattering or transmission of electrons.
3	Lenses used	Lenses are made of glass.	Lenses are made of electromagnets.
4	Specimen type	Fixed or unfixed, stained or unstained, living or non-living.	Fixed, stained, and non-living.
5	Specimen preparation	Less tedious and simple.	It generally involves harsher processes, e.g. using corrosive chemicals. More skill required – both to prepare specimens and to interpret EM images (due to artifacts).
6	Magnification power	Low magnification of up to 1,500x.	High magnification of up to 1,000,000x.
7	Resolving power	Low resolving power, usually below 0.30μm.	The high resolving power of up to 0.001μm, about 250 times higher than the light microscope.
8	Types/ Variants	• Dark-field microscope • Phase-contrast microscope • Fluorescent microscope • Confocal microscope • Polarized microscope • Differential interference contrast microscope	• Transmission electron microscope (TEM) • Scanning electron microscope (SEM)

Ecological aspects of microorganisms

- Ecology studies the interaction of organisms in an area with the surrounding environment. This interaction constitutes an overall adaptation of the organisms to their environment, including the continuity of species. These interactions can be **inter-specific** (interactions with different species) or **intra-specific** (interactions between the same species). There are five types of interactions between other species, as listed below:

1. **Competition and Predation:** When one entity hunts another animal to suffice its nutritional requirements, it is referred to as predation. A predator is an entity that hunts its prey. For example, a snake eats a frog. Here snake is the predator, and the frog is its prey. Competition, on the other hand, is when populations or even an individual compete for food resources. It is often referred to as exploitative or consumptive competition. For example, competition for territory.
2. **Commensalism:** It is an imbalanced type of interaction wherein one entity benefits while the other is neither harmed nor benefited. There are four types of commensal associations.

- **Inquilinism –** An entity occupies the living habitat of another species (burrow, nest)
- **Chemical commensalism –** A bacteria produces a chemical that nurtures another bacteria
- **Phoresy –** An organism tentatively attaches itself to another entity for transportation requirements.
- **Metabiosis –** One entity is dependent on the other for survival

Ecological aspects of microorganisms

3. Parasitism: One entity benefits from other entities and is harmed but not necessarily killed. The entity that is harmed is the **host**, and the one benefit is the **parasite**. When the host is killed, this type of behavior is referred to as **parasitoidism**. These parasites can live on the surface of the host, often addressed as **ectoparasites** (fleas, leeches), while **endoparasites** live inside the host. Endoparasites can be subdivided into intracellular parasites (live inside cells) and intercellular parasites (live in spaces between cells).

4. Mutualism: Both species involved in the interaction are benefited. These interactions take place in three patterns.

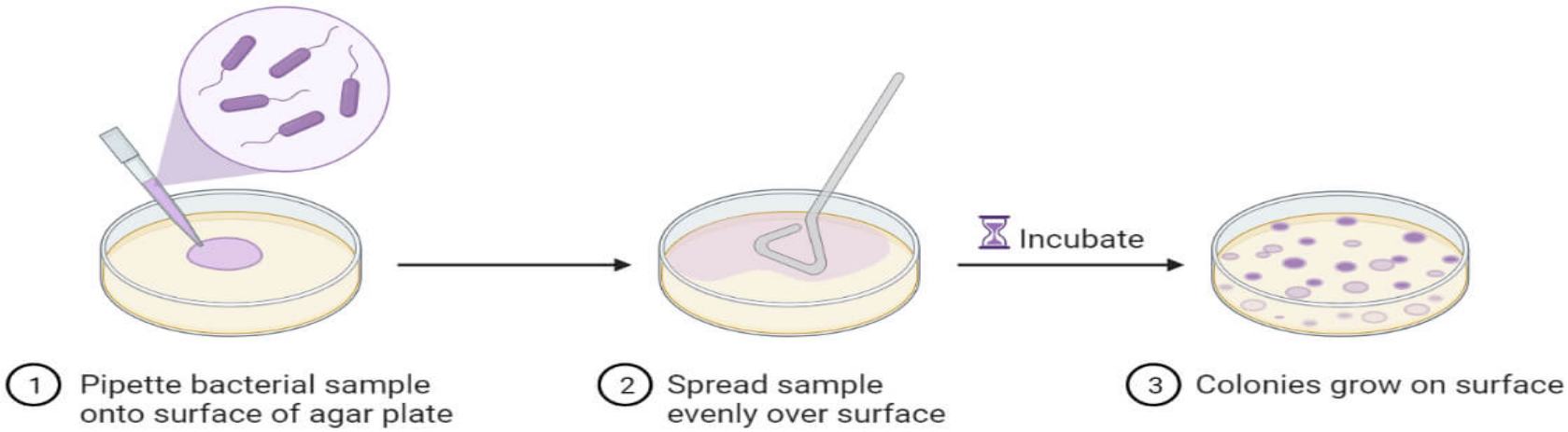
- **Facultative mutualism –** Species survive on their own under favorable conditions
- **Obligate mutualism –** One species is dependent for survival on the other
- **Diffusive mutualism –** One entity can live with multiple partners

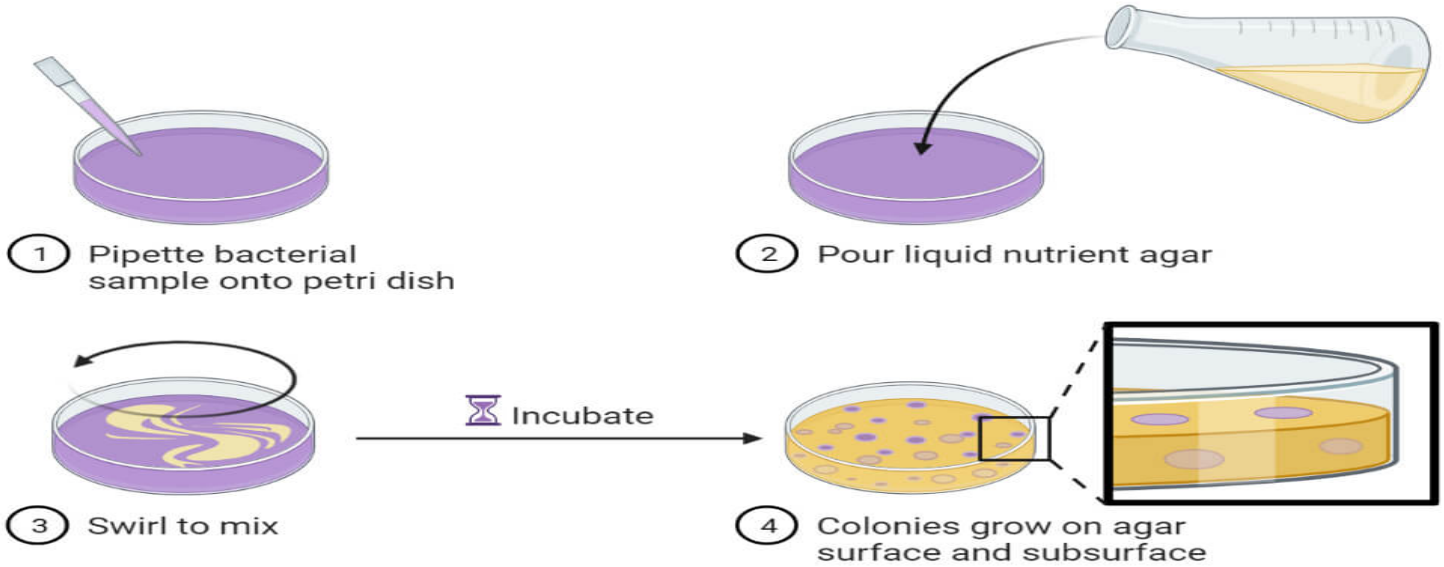
5. Amensalism: In this type of interaction, when one population finds itself in danger, the other population is not majorly affected. For instance, Tall and wide plants hinder the growth of comparatively smaller plants. Some plants even secrete substances that repress the growth of nearby plants to remove competition.

Microbial Culture

- In a laboratory, microorganisms are cultivated in a growth environment containing the required nutrients called a **medium**. Microorganisms growing in or on such a medium form a **culture**.
- If more than one type of organism is present, it is referred to as a **mixed culture**. If only one type of organism is present, it is referred to as a **pure culture**.
- **Steps of microbial culture:**

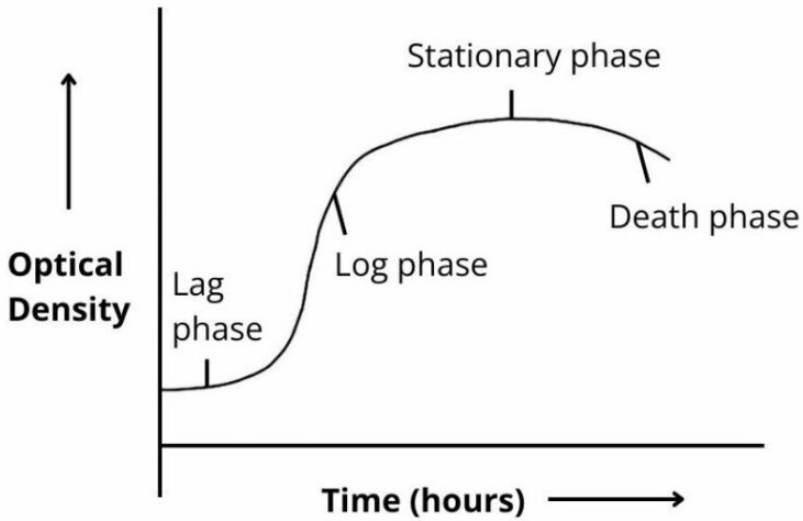
1. Inoculation: The term "inoculum" refers to the tiny sample transmitted from the culture.
2. Isolation: A particular microbial strain can be isolated using the microbiological technique of isolation by cultivating the microbes in a specific culture medium. The process must be performed numerous times to completely purify the microbial strain and generate a pure culture.
3. Spread plate method: The spread plate method can be used to count the number of bacteria present in a sample. The spread plate technique is most frequently utilized to isolate and identify a variety of microbial flora present in environmental samples like soil, food, or any other materials. (Top pic in next slide)
4. Pour plate method: It is done to gauge the number of live bacteria in a microbiological suspension. (Bottom pic next slide)


Incubate
1 Pipette bacterial sample onto surface of agar plate
2 Spread sample evenly over surface
3 Colonies grow on surface


1 Pipette bacterial sample onto petri dish
2 Pour liquid nutrient agar
Incubate
3 Swirl to mix
4 Colonies grow on agar surface and subsurface

Bacterial Growth Kinetics

- Bacterial growth is associated with many complex anabolic and catabolic processes. When microbes receive nutrients and the necessary environmental conditions, they become metabolically active and proliferate.
- Development occurs on two levels. On one level, a cell synthesizes new cell components and grows in size, while on another level, the population of cells rises.
- Throughout their development cycles, bacteria reproduce several times resulting in a substantial increase in population size. Bacteria reproduce **asexually** through a mechanism known as **binary fission**.
- During this process, chromosomal DNA replicates, and then the bacterial membrane and cell wall grow inward to meet and divide the cell. The two cells separate, concluding the process.



Bacterial Growth Kinetics

- **Lag phase**: When microorganisms are introduced into fresh growth media, they do not directly start to reproduce. It is a phase where cells are adjusting to the new growth conditions, and so also called the adaptation period. In the lag phase, bacteria are not metabolically inactive but repair themselves, produce enzymes, and synthesize chemicals required for carrying out physiological and biochemical processes. It is a phase where bacteria prepare themselves for cell division and multiplication.
- **Exponential or log phase**: Exponential phase is a period of rapid cell division. Once the cell has accumulated all the necessary components for cell division, it starts diving exponentially. Microorganisms develop and divide at the fastest pace feasible given their genetic capacity, the type of medium, and the ambient conditions. The exponential or log phase of growth is characterized by predictable population doublings, where 1 cell becomes 2 cells, 4 cells become 8, and so on. During the exponential phase, their rate of growth is constant; that is, they complete the cell cycle and double in number at regular intervals. Situations that are optimum for the cells result in very rapid growth (and a steeper slope on the growth curve), whereas conditions that are less than ideal result in slower growth. Because cells at the exponential phase of growth are the healthiest and most uniform, they are used in the majority of experiments. This phase's length depends on bacterial circumstances.
- **Stationary phase**: Throughout the exponential phase, organisms utilize the nutrients for their growth. Therefore, the nutrients in this closed culture system deplete at some point, waste products accumulate, and even space may run out, reducing cells. The growth curve flattens. In the stationary phase, the number of cells reproducing equals the number of cells dying. Some microorganisms produce secondary metabolites like antibiotics in this phase.
- **Death phase**: The organisms cannot stay in the stationary phase infinitely. As conditions no longer encourage growth, the condition of the cells deteriorates, causing the growth curve to decrease.

Thank you.